DSCI 552 Homework 5

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AI Usage Disclosure

I used AI tools to help me learn the relevant programming techniques and machine learning concepts, understand code implementations, and assist with debugging.

Requirements and experimental protocol

The two-page assignment and all five footnotes were read, as were both supplied .names files and the supplied KDnuggets article. Local data are used without downloading replacements. The original question order is retained: 1(a) data, 1(b) full tree, 1(c) all rules, 1(d) pruning; 2(a) ordered split, 2(b) imputation, 2(c) correlations, 2(d) CV, 2(e) 11 feature plots, 2(f) OLS, 2(g) Ridge, 2(h) both LASSOs, 2(i) PCR, 2(j) XGBoost.

Seed 552 is fixed. All regression models retain all 122 predictive variables as candidate inputs; the 11-variable CV ranking is descriptive only. Regression tuning uses the same shuffled five-fold KFold within the training portion, minimizing validation MSE. The mandated external split is never shuffled. Imputation, scaling and PCA are fitted separately in each training fold through Pipeline. The test set is evaluated once per final fitted model, after all six models are frozen. No test metric is used to revise a model.

Report guide. This report covers all 14 subquestions within the instructor's inclusive 20-page limit (cover, references and any appendices included). Detailed tables and enlarged diagnostic plots are retained in the executed Notebook, results/ and figures/. The compact figure views below use the same saved values and original training records; no models are fitted by this report generator.

Execution environment: Python 3.13.5; numpy 2.5.4; pandas 3.1.0; scikit-learn 1.9.1; xgboost 3.4.1; matplotlib 3.11.2. Full versions: results/versions.csv and requirements.txt.

1(a) Acute Inflammations: data and label powerset

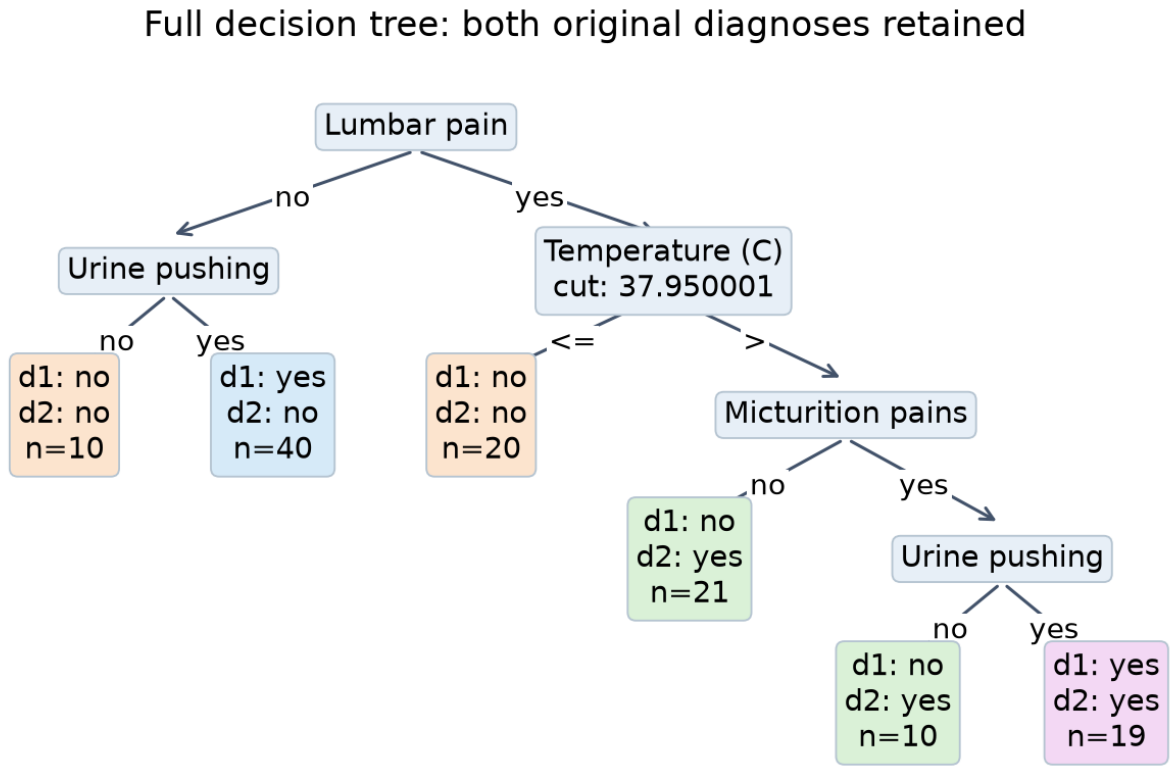
The actual byte encoding is detected from the BOM (UTF-16 if present, otherwise UTF-8); this takes precedence over the ASCII statement in diagnosis.names. Tabs separate fields. Decimal commas in temperature are converted to decimal points. The five binary symptoms and two diagnoses map no to 0 and yes to 1. The label powerset code is 2*d1+d2, so both original diagnoses remain identifiable. d1 denotes inflammation of urinary bladder; d2 denotes nephritis of renal pelvis origin. These are expert-created illustrative records, not evidence of clinical performance.

Powerset class	Bladder inflammation (d1)	Renal pelvis nephritis (d2)	Records
0	no	no	30
1	no	yes	31
2	yes	no	40
3	yes	yes	19

Validated: 120 rows, 6 predictors, 2 labels; encoding **utf-16**; no missing values.

1(b) Full-data decision tree

A CART classifier uses Gini impurity and the powerset target, with ccp_alpha=0 and otherwise unrestricted growth. As requested, the final tree is fitted to all 120 records. Tree class labels explicitly show both diagnoses (0=no, 1=yes). Training accuracy is a description of fit, not a held-out estimate.



1(b). Complete saved tree topology and leaf predictions. d1 = bladder inflammation; d2 = renal-pelvis nephritis; n = leaf support. Binary edges are symptom no/yes; temperature edges compare to the displayed cut. Exact thresholds and every path appear in 1(c); the original impurity/count diagram remains in Notebook 1(b).

1(c) All root-to-leaf IF-THEN rules

Each conjunction is one complete path. Binary branches are translated back to yes/no, and leaf classes back to the original two diagnoses. Temperature thresholds retain the model threshold to sufficient precision. n is the training support of the leaf. The executable path conditions are checked against tree predictions for every record, with exactly one matching rule per record.

Rule 1. IF Lumbar pain = no AND Urine pushing = no THEN bladder inflammation = no; renal pelvis nephritis = no (n=10).

Rule 2. IF Lumbar pain = no AND Urine pushing = yes THEN bladder inflammation = yes; renal pelvis nephritis = no (n=40).

Rule 3. IF Lumbar pain = yes AND Temperature (C) <= 37.950001 THEN bladder inflammation = no; renal pelvis nephritis = no (n=20).

Rule 4. IF Lumbar pain = yes AND Temperature (C) > 37.950001 AND Micturition pains = no THEN bladder inflammation = no; renal pelvis nephritis = yes (n=21).

Rule 5. IF Lumbar pain = yes AND Temperature (C) > 37.950001 AND Micturition pains = yes AND Urine pushing = no THEN bladder inflammation = no; renal pelvis nephritis = yes (n=10).

Rule 6. IF Lumbar pain = yes AND Temperature (C) > 37.950001 AND Micturition pains = yes AND Urine pushing = yes THEN bladder inflammation = yes; renal pelvis nephritis = yes (n=19).

1(d) Cost-complexity pruning and interpretable minimum

CART minimizes training leaf impurity plus alpha times the number of leaves. To define a useful minimum, a predeclared grid of 401 alpha values from 0 through 0.4 is evaluated by shuffled stratified five-fold CV on exact-match powerset accuracy. This grid is independent of label-derived pruning breakpoints. Select the largest alpha whose mean accuracy is within one standard error of the best mean; ties therefore favor fewer leaves. The standard error is the sample standard deviation of five fold accuracies divided by sqrt(5). This is a practical simplicity rule, not a proof of globally minimal trees or statistical equivalence. Final candidates are then refitted on all 120 records. The full-data pruning path is reported descriptively, not used to construct the CV grid. A stump is explicitly compared to the chosen model.

alpha breakpoint	Leaf impurity	Nodes	Leaves	Depth	Train exact accuracy
0	0	11	6	4	1
0.09816667	0.1963333	7	4	2	0.8416667
0.1333333	0.3296667	5	3	2	0.7583333
0.182	0.5116667	3	2	1	0.5916667
0.2229167	0.7345833	1	1	0	0.3333333

Tree	ccp_alpha	Leaves	Depth	Train acc.	CV acc.
Unpruned	0	6	4	1	0.9916667
Selected	0.089	6	4	1	0.9916667
Reduced path candidate	0.09816667	4	2	0.8416667	N/A
Stump baseline	1	1	0	0.3333333	0.3333333

N/A denotes the descriptive path candidate, which is not a replacement CV selection. Full Hamming-loss and error-count details remain in results/tree_comparison.csv and results/tree_reduced_candidate_comparison.csv.

The best CV accuracy is **0.9917**, SE **0.0083**; the acceptance threshold is **0.9833**. Selected alpha = **0.089**. The selected tree has **6 leaves** and depth **4** versus **6 leaves** and depth **4** before pruning. No smaller subtree meets this criterion; keeping the original structure is preferable to sacrificing useful accuracy.

These CV scores informed alpha selection and are not an independent post-selection performance estimate. There is no separate test set in Question 1, in accordance with the full-data fitting requirement.

The original diagnostic plot is retained in the Notebook and figures/pruning_selection.png; corresponding numeric tables remain in results/.

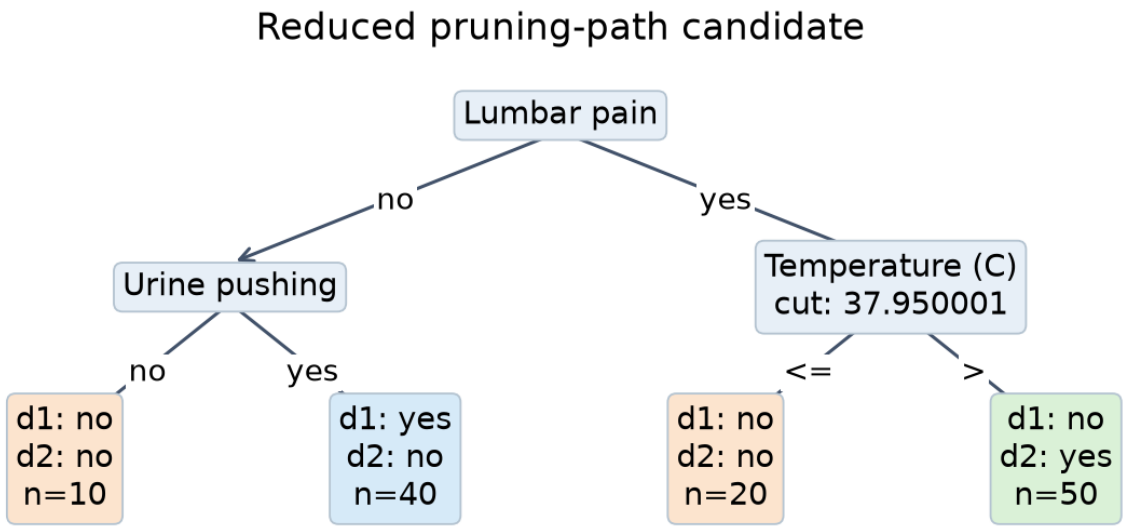
One-SE tree and rules: the selected tree has exactly the same topology and all 6 rules as 1(b)-1(c). Thus Rules 1-6 are also the complete one-SE rule set. Its separate original diagram and rule file remain in figures/tree_pruned.* and results/rules_pruned.json.

Additional 1(d): an actually smaller pruning-path candidate. The original CV results and one-SE selection above are retained. For an interpretable alternative, fit the trees at the full-data cost-complexity breakpoints and choose the first non-stump subtree whose leaf count is strictly below the unpruned tree. This rule does not prescribe a leaf count or use the regression test data. The alternative is a descriptive complexity/accuracy comparison, not a replacement for the CV-selected model.

The first actual reduction occurs at **ccp_alpha = 0.0981666666667**. It changes the tree from **6 leaves, depth 4, training exact-match accuracy 100.00%** to **4 leaves, depth 2, accuracy 84.17%**: a loss of **15.83 percentage points** and **19 / 120** misclassified diagnosis pairs. All figures and metrics here are computed from fitted trees, rather than a hard-coded leaf target.

The reduced tree omits these splitting features: **Micturition pains**. It gives **4** diagnostic paths and at most **2** decisions per path, compared with **6** paths and at most **4** decisions before pruning. This is easier to trace, but merged leaves assign a majority diagnosis pair to patients whose original diagnoses differ. The errors comprise **19 bladder-inflammation labels** and **0 renal-pelvis-nephritis labels**. Training accuracy describes these illustrative records; it is not a generalization estimate.

Selection conclusion is unchanged: the original one-SE procedure retains **ccp_alpha = 0.089**, with **6 leaves**. The smaller candidate is shown to make the interpretability cost concrete, but its loss of diagnostic accuracy is not hidden or accepted as a new CV optimum. There is no claim that this full-data path choice has an independent validation estimate. A stump would be still shorter, but would discard substantially more diagnostic information.



1(d). Actually reduced candidate reconstructed from its saved rules. Both diagnosis predictions and every terminal leaf are shown; all rules follow.

Reduced-candidate rule 1. IF Lumbar pain = no AND Urine pushing = no THEN bladder inflammation = no; renal pelvis nephritis = no (n=10).

Reduced-candidate rule 2. IF Lumbar pain = no AND Urine pushing = yes THEN bladder inflammation = yes; renal pelvis nephritis = no (n=40).

Reduced-candidate rule 3. IF Lumbar pain = yes AND Temperature (C) <= 37.950001 THEN bladder inflammation = no; renal pelvis nephritis = no (n=20).

Reduced-candidate rule 4. IF Lumbar pain = yes AND Temperature (C) > 37.950001 THEN bladder inflammation = no; renal pelvis nephritis = yes (n=50).

2(a) Communities and Crime: data and ordered split

Read the 128 attribute names directly from @attribute lines in communities.names. Keep original file order: training is original rows 1-1495, test is rows 1496-1994. The target is ViolentCrimesPerPop. The provided data were already normalized approximately to [0,1]; this is different from standardization to zero mean and unit variance. The data combine the 1990 Census, 1990 LEMAS and 1995 FBI UCR; normalization parameters used by the original dataset creator cannot be undone. All additional preprocessing here is learned solely from training data.

Validated raw shape: **(1994, 128)**. Training: **(1495, 122)**. Test: **(499, 122)**. Target excluded from predictors; all five nonpredictive fields removed.

2(b) Missing values and imputation

Question marks are parsed as NaN. Median imputation is robust to skew and retains all 122 predictors, including sparsely observed LEMAS variables. A median cannot reconstruct missing information or remove selection bias: LEMAS coverage is highly nonrandom according to the supplied documentation. No rows or predictive columns are dropped. An all-missing column within a training fold is retained and filled with zero via keep_empty_features=True (the audit below checks whether this fallback is actually needed). The descriptive training matrix is median-imputed using training rows only. Predictive CV uses a separate SimpleImputer inside every Pipeline; the descriptive matrix is never passed to GridSearchCV.

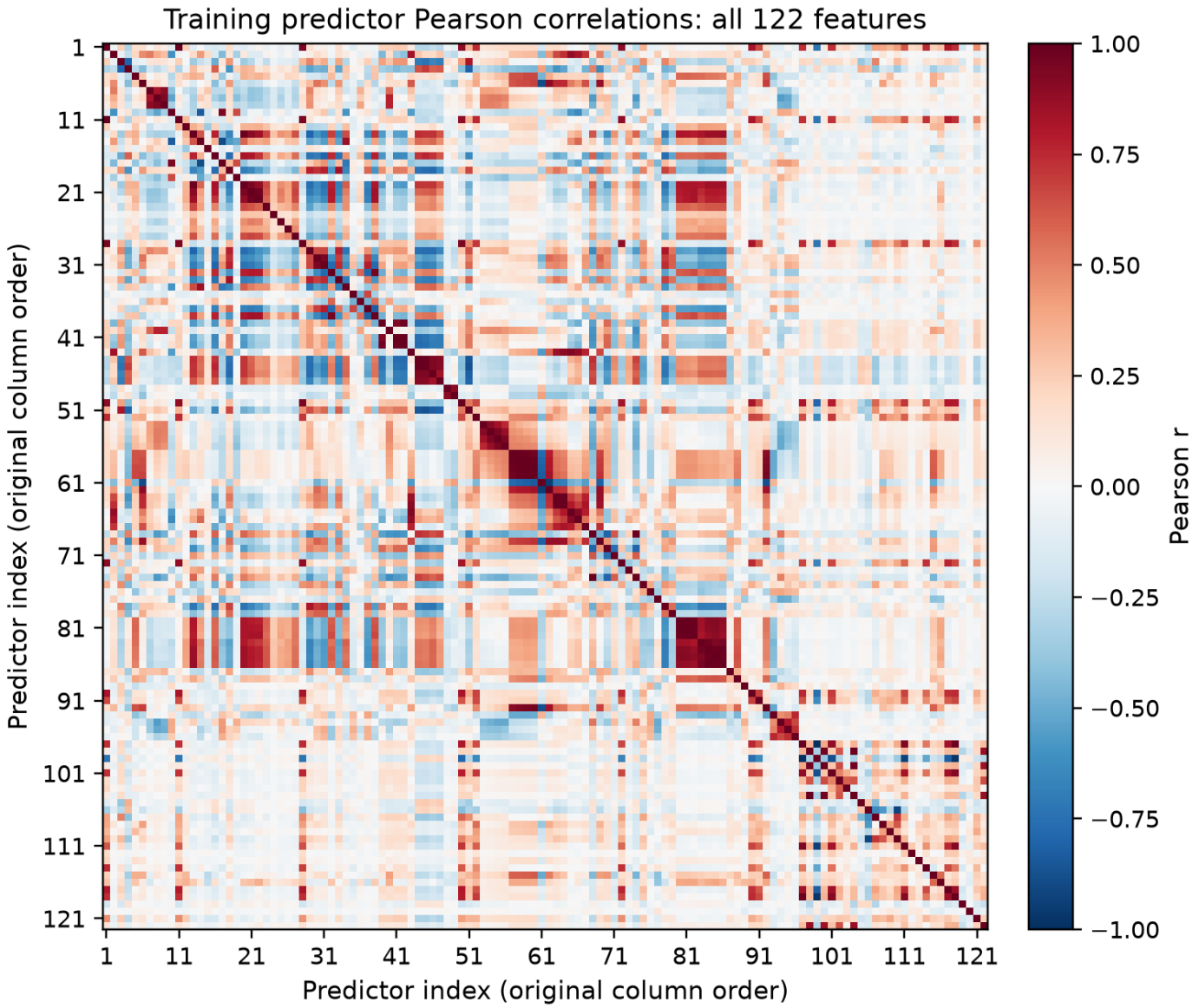
Predictors	Missing training rows	Fraction
1	1	0.0006689
22	1256	0.8401338

Feature-by-feature missingness remains in results/missingness.csv and Notebook 2(b). The group counts above summarize the same saved table.

All-missing training-fold columns across five folds: **0**. Number of predictive columns dropped: **0**.

2(c) Feature correlation matrix

Pearson correlations use all 122 median-imputed training predictors; the target is excluded. The complete matrix is shown with readable predictor indices in the original predictive-column order. The index-to-name key is results/correlation_feature_index.csv. The fully named matrix, four enlarged panels and exact values remain in the Notebook, figures/correlation_overview.*, figures/correlation_panel_1.* through correlation_panel_4.*, and results/feature_correlations.csv. Strong correlated blocks suggest redundancy; imputation may alter correlations, especially for sparse LEMAS fields. Association does not establish causation.



2(c). Full 122 x 122 correlation matrix. Indices correspond to results/correlation_feature_index.csv; color scale is shared across every cell.

2(d) Coefficient of variation for every predictor

CV = s/m uses sample standard deviation with ddof=1 and sample mean over the median-imputed training rows only. CV is computed before standardization. A nonzero constant with nonzero mean has CV=0; any zero (or numerically near-zero, absolute tolerance 1e-12) mean has undefined CV and is excluded from ranking. All-zero columns are also undefined. This rule avoids replacing an undefined ratio with an arbitrary large number. The target and identifier fields are excluded. CV is sensitive to the dataset's normalization origin and does not measure predictive importance.

Complete CV table: all 122 predictor means, sample SDs and CVs are retained in the Notebook under 2(d) and results/coefficient_of_variation.csv. The ratio calculation and edge-case treatment are unchanged.

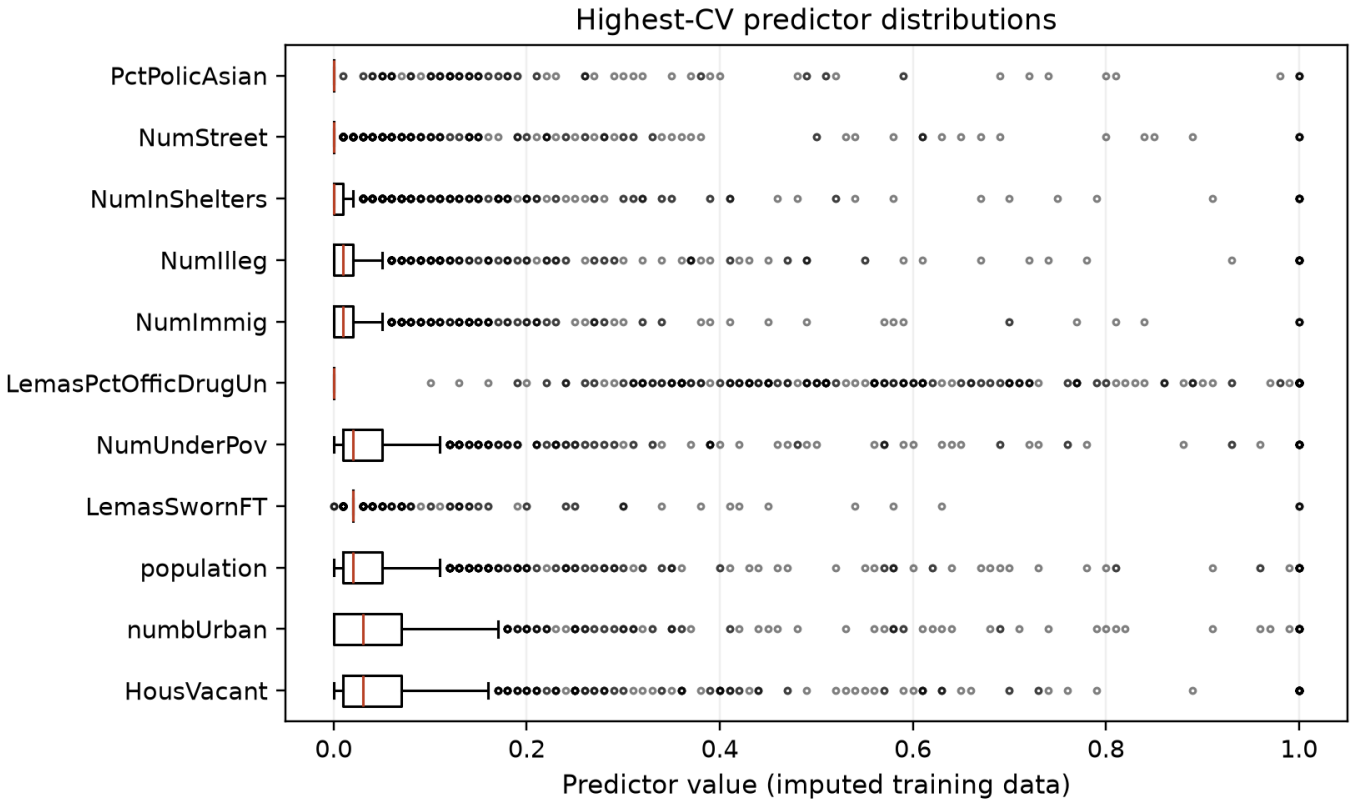
Undefined CVs: **0**; constant training predictors: **0**. All 122 coefficients remain in the Notebook and results/coefficient_of_variation.csv; the 11 highest are included below.

2(e) The 11 highest-CV predictors: scatter plots and box plots

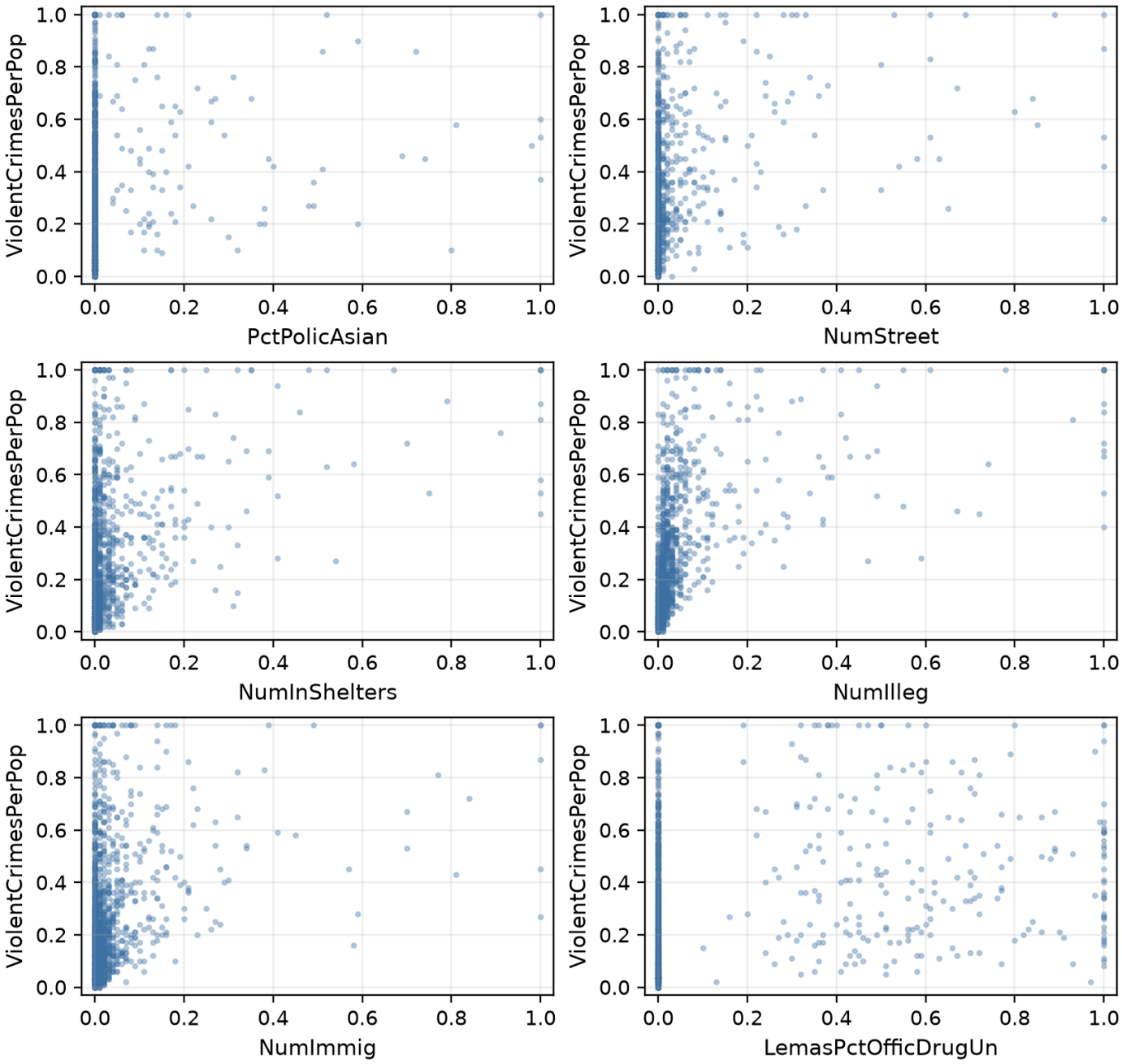
The count is floor(sqrt(128))=11 as specified, even though only 122 columns are predictive. These plots use the same imputed training rows. Each scatter plot has a predictor on the horizontal axis and ViolentCrimesPerPop on the vertical axis; each box plot summarizes one predictor. The 11-feature selection does not restrict subsequent regression models.

Scatter patterns can suggest skew, outliers, nonlinear associations, and heteroskedasticity. Neither scatter plots nor box plots establish statistical significance. Formal inference would require a specified null, a suitable model and assumptions, uncertainty estimates and multiple-comparison handling. Large CV alone does not imply a significant relationship with the target, and these observational data cannot establish causal effects.

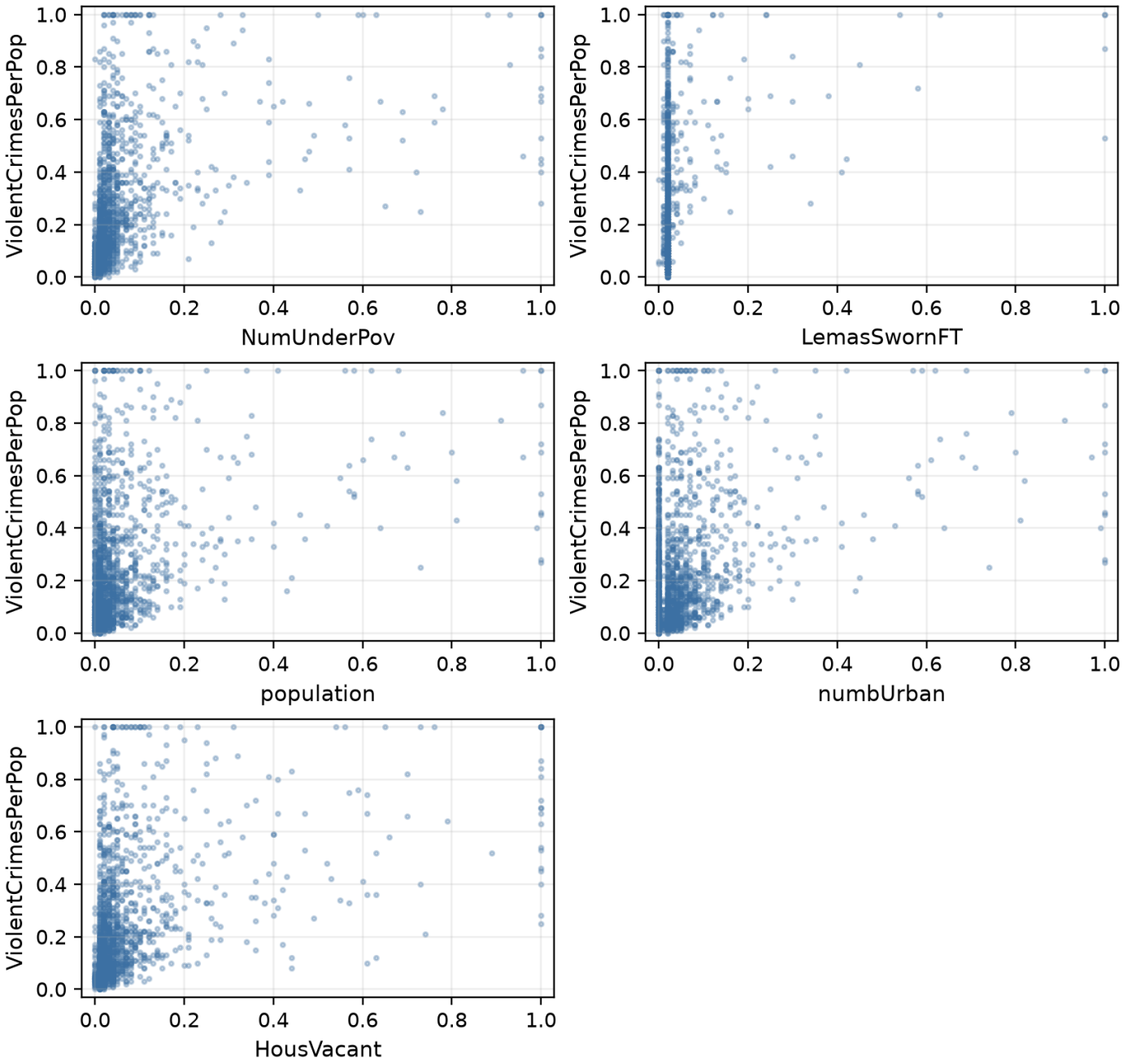
Feature	Mean	Sample SD (ddof=1)	CV
PctPolicAsian	0.01625418	0.08983781	5.527059
NumStreet	0.02298328	0.09866544	4.292923
NumInShelters	0.02959866	0.1027355	3.470952
NumIlleg	0.03720401	0.1138058	3.058964
NumImmig	0.03059532	0.08954134	2.926635
LemasPctOfficDrugUn	0.09330435	0.2382009	2.552946
NumUnderPov	0.0568495	0.1331667	2.342443
LemasSwornFT	0.02874916	0.06462932	2.248042
population	0.05816722	0.1303588	2.241105
numbUrban	0.06464883	0.1317841	2.038461
HousVacant	0.07647492	0.1505384	1.968467



2(e). One box per selected predictor, on a common numeric scale; medians, interquartile ranges, whiskers and outliers remain visible.



2(e). Scatter plots, panel 1/2. Every predictor is on x; the crime target is on y. Same imputed training observations as the original plots.



2(e). Scatter plots, panel 2/2. Every predictor is on x; the crime target is on y. Same imputed training observations as the original plots.

2(f) Ordinary least squares

OLS fits an intercept and all 122 imputed predictors by minimizing residual sum of squares. It has no tuned hyperparameter. Coefficients can be unstable under multicollinearity; the matrix rank and condition number below describe this concern. Main test error is MSE = mean((y-yhat)^2), with RMSE=sqrt(MSE) and R-squared also reported. All model test results are deliberately collected together after Question 2(j) has frozen every fitted model.

Centered training design rank = **121 / 122**; condition number = **3.51e+16**.

OLS: test MSE = 0.01796622; RMSE = 0.13403811; R-squared = 0.622204.

2(g) Ridge regression

A Pipeline applies median imputation and StandardScaler before Ridge. Scikit-learn Ridge minimizes ||y-Xb||^2 + alpha*||b||^2, with an unpenalized intercept. Thus its alpha is the assignment's lambda under this convention. A 49-point logarithmic grid from 1e-6 through 1e6 is searched using training-fold MSE; if an artificial endpoint wins, the range is expanded. Standardizing equalizes penalty treatment across features, despite their existing [0,1] normalization.

Ridge: model__alpha = **56.23413251903491**; five-fold CV MSE = **0.01914446**; final grid [1e-06, 1000000.0], 49 candidates. Convergence warnings: **0**. The selected value is interior to the final search range.

Ridge: test MSE = 0.01747522; RMSE = 0.13219389; R-squared = 0.632528.

2(h) LASSO without and with standardization

Both models use median imputation inside CV. The first retains the original supplied normalized units; the second applies fold-fitted StandardScaler. Lasso minimizes ||y-Xb||^2/(2*n) + alpha*||b||_1 with an unpenalized intercept, so numerical alpha is not directly comparable with Ridge alpha. Each model searches 41 logarithmically spaced values from 1e-5 to 1e-1, expanding artificial boundaries if needed. A strict tolerance of 1e-7 and max_iter=200000 are used; convergence warnings cause execution to fail instead of being hidden. Selected variables have absolute fitted coefficient greater than 1e-10. Coefficients for the standardized fit are expressed per training standard deviation; original-unit equivalents are also exported.

LASSO_unscaled: model__alpha = **0.0001**; five-fold CV MSE = **0.01899094**; final grid [1e-05, 0.1], 41 candidates. Convergence warnings: **0**. The selected value is interior to the final search range.

LASSO_unscaled: all 75 selected variables (read each column from top to bottom).

racepctblack	PctLess9thGrade	PersPerOccupHous	NumStreet
racePctAsian	PctBSorMore	PctPersOwnOccup	PctForeignBorn
agePct12t21	PctEmploy	PctPersDenseHous	PctSameCity85
agePct12t29	PctEmplManu	PctHousLess3BR	LemasSwFTFieldPerPop
agePct65up	PctOccupManu	MedNumBR	LemasTotalReq
pctUrban	MalePctDivorce	HousVacant	PolicReqPerOffic
pctWWage	MalePctNevMarr	PctHousOccup	RacialMatchCommPol
pctWFarmSelf	FemalePctDiv	PctVacantBoarded	PctPolicBlack
pctWInvInc	PctKids2Par	PctVacMore6Mos	PctPolicHisp
pctWSocSec	PctYoungKids2Par	MedYrHousBuilt	PctPolicAsian
pctWPubAsst	PctTeen2Par	PctHousNoPhone	NumKindsDrugsSeiz
pctWRetire	PctWorkMom	PctWOFullPlumb	PolicAveOTWorked
whitePerCap	NumIlleg	OwnOccLowQuart	LandArea
blackPerCap	PctIlleg	RentLowQ	PopDens
indianPerCap	NumImmig	MedRent	PctUsePubTrans
AsianPerCap	PctImmigRec10	MedRentPctHousInc	PolicCars
OtherPerCap	PctRecImmig10	MedOwnCostPctInc	LemasPctPolicOnPatr
HispPerCap	PctNotSpeakEnglWell	MedOwnCostPctIncNoMtg	LemasGangUnitDeploy
PctPopUnderPov	PctLargHouseOccup	NumInShelters	

All coefficients, including original-unit equivalents, remain in results/coefficients_LASSO_unscaled.csv; the selected-variable table is results/selected_LASSO_unscaled.csv and Notebook 2(h).

LASSO_unscaled selects 75 variables, all listed above. Final solver iterations: 573; dual gap: 5.54e-09.

LASSO_scaled: model__alpha = **0.000630957344480193**; five-fold CV MSE = **0.01910257**; final grid [1e-05, 0.1], 41 candidates. Convergence warnings: **0**. The selected value is interior to the final search range.

LASSO_scaled: all 76 selected variables (read each column from top to bottom).

racepctblack	PctEmploy	PersPerOwnOccHous	NumStreet
racePctAsian	PctEmplManu	PctPersOwnOccup	PctForeignBorn
agePct12t21	PctOccupManu	PctPersDenseHous	PctSameCity85
agePct12t29	MalePctDivorce	PctHousLess3BR	LemasSwFTFieldPerPop
pctUrban	MalePctNevMarr	MedNumBR	LemasTotalReq
pctWWage	FemalePctDiv	HousVacant	PolicReqPerOffic
pctWFarmSelf	PctKids2Par	PctHousOccup	RacialMatchCommPol
pctWInvInc	PctYoungKids2Par	PctVacantBoarded	PctPolicBlack
pctWSocSec	PctTeen2Par	PctVacMore6Mos	PctPolicHisp
pctWPubAsst	PctWorkMom	MedYrHousBuilt	PctPolicAsian
pctWRetire	NumIlleg	PctHousNoPhone	OfficAssgnDrugUnits
whitePerCap	PctIlleg	PctWOFullPlumb	NumKindsDrugsSeiz
blackPerCap	NumImmig	OwnOccLowQuart	PolicAveOTWorked
indianPerCap	PctImmigRecent	RentLowQ	LandArea
AsianPerCap	PctImmigRec10	MedRent	PopDens
OtherPerCap	PctRecImmig10	MedRentPctHousInc	PctUsePubTrans
HispPerCap	PctNotSpeakEnglWell	MedOwnCostPctInc	PolicCars
PctPopUnderPov	PctLargHouseOccup	MedOwnCostPctIncNoMtg	LemasPctPolicOnPatr
PctLess9thGrade	PersPerOccupHous	NumInShelters	LemasGangUnitDeploy

All coefficients, including original-unit equivalents, remain in results/coefficients_LASSO_scaled.csv; the selected-variable table is results/selected_LASSO_scaled.csv and Notebook 2(h).

LASSO_scaled selects 76 variables, all listed above. Final solver iterations: 546; dual gap: 4.74e-09.

Standardization changes the effective L1 penalty in original units: a unit change in a low-variance feature is treated differently after scaling. Hence the fitted supports and optimal penalties can differ. Correlated predictors may substitute for each other, so an excluded variable is not necessarily irrelevant. The test-error comparison follows in the frozen evaluation table.

LASSO_unscaled: test MSE = 0.01765779; RMSE = 0.13288261; R-squared = 0.628689.

LASSO_scaled: test MSE = 0.01772176; RMSE = 0.13312311; R-squared = 0.627344.

2(i) Principal component regression (PCR)

The Pipeline is median imputation -> standardization -> full-SVD PCA -> ordinary least squares. M is selected by the same five training folds and validation MSE. All integer values 1 through 122 are searched, so neither endpoint is an artificial truncated search boundary. PCA directions, imputation medians and scaling statistics are refitted within each fold. PCA maximizes predictor variance, not association with the response; a high explained-variance fraction alone does not determine the best M.

PCR: pca__n_components = **93**; five-fold CV MSE = **0.01944865**; final grid [1, 122], 122 candidates. Convergence warnings: **0**. The selected value is interior to the final search range.

The selected 93 components explain **99.7959%** of standardized training predictor variance.

PCR: test MSE = 0.01826425; RMSE = 0.13514528; R-squared = 0.615937.

2(j) L1-penalized boosting with XGBoost

Implementation assumption: interpret the allowed XGBoost option as **booster="gbtree"** with a CV-selected **reg_alpha**, an L1 penalty on scalar leaf weights. Standard tree leaves predict constants; they do not fit LASSO regressions at nodes. This differs from a literal tree with local multivariate L1 regressions. The standard gbtree implementation fulfills the explicit XGBoost option under this stated interpretation; no unsupported local-linear-tree claim is made. gblinear would fit a linear booster and would not be a boosting tree.

For a fixed tree structure, the regularizer includes reg_alpha*sum(|w_j|) + (reg_lambda/2)*sum(w_j^2). The L1 term can shrink leaf weights to zero and affects split gains. It is not direct coefficient selection over the original 122 features. See the official XGBoost parameter documentation and boosted-tree introduction.

The fixed, predeclared configuration uses squared-error loss, 400 trees, learning_rate=0.04, max_depth=3, min_child_weight=5, reg_lambda=1, subsample=1, colsample_bytree=1, gamma=0, hist trees, n_jobs=1 and seed 552. Only reg_alpha is tuned: zero plus 29 positive log-spaced candidates from 1e-5 through 1e2, expanding if necessary. Median imputation remains inside the Pipeline for consistent data handling. No test-set early stopping is used; other tree settings are fixed modeling assumptions, not claimed globally optimal.

XGBoost: model__reg_alpha = **0.00017782794100389227**; five-fold CV MSE = **0.01970332**; final grid [0.0, 100.0], 30 candidates. Convergence warnings: **0**. The selected value is interior to the final search range.

Model	Selected parameter	CV MSE
Ridge	alpha = 56.23413	0.01914446
LASSO_unscaled	alpha = 0.0001	0.01899094
LASSO_scaled	alpha = 0.000631	0.01910257
PCR	n_components = 93	0.01944865
XGBoost	reg_alpha = 0.0001778	0.01970332

Every selected setting is interior to its admissible search range; no convergence warnings occurred. Full grid bounds, candidates and fold scores: results/tuning_summary.csv, results/search_audit_*.json and results/cv_*.csv.

Model	Test MSE	RMSE	R-squared	LASSO count
OLS	0.01796622	0.13403811	0.622204	N/A
Ridge	0.01747522	0.13219389	0.632528	N/A
LASSO_unscaled	0.01765779	0.13288261	0.628689	75
LASSO_scaled	0.01772176	0.13312311	0.627344	76
PCR	0.01826425	0.13514528	0.615937	N/A
XGBoost	0.01621407	0.12733449	0.659048	N/A

OLS: test MSE = 0.01796622; RMSE = 0.13403811; R-squared = 0.622204.

Ridge: test MSE = 0.01747522; RMSE = 0.13219389; R-squared = 0.632528.

LASSO_unscaled: test MSE = 0.01765779; RMSE = 0.13288261; R-squared = 0.628689.

LASSO_scaled: test MSE = 0.01772176; RMSE = 0.13312311; R-squared = 0.627344.

PCR: test MSE = 0.01826425; RMSE = 0.13514528; R-squared = 0.615937.

XGBoost: test MSE = 0.01621407; RMSE = 0.12733449; R-squared = 0.659048.

Standardized LASSO selected **76** variables versus **75** without standardization. Its test MSE is **0.00006398 higher**. This is a final held-out comparison, not a reason to retune. Lowest observed test MSE among the prespecified fits: **XGBoost**. This descriptive ranking on one fixed split does not establish a statistically reliable

winner or estimate the performance of a newly selected model.

The original diagnostic plot is retained in the Notebook and figures/model_test_mse.png; corresponding numeric tables remain in results/.

The original diagnostic plot is retained in the Notebook and figures/cv_curves.png; corresponding numeric tables remain in results/.

Reproducibility, validation and limitations

The notebook must run from the project root. All input and output paths are relative. Pinned dependencies, raw data hashes, fold assignments, full search results, LASSO coefficient tables, rules and test predictions are included. Every code cell is executed in order by a fresh Jupyter kernel using scripts/run_notebook.py. scripts/make_report.py creates the English PDF from the same recorded narrative, tables and figures, so the report is tied to computed outputs.

The original nonrandom external split may have distribution shifts; shuffled internal CV assumes exchangeability within the training portion. It does not account for possible geographic or social dependence. Median filling of heavily missing police variables is an information limitation, not a claim that missingness is ignorable. The 11 highest-CV features are not a formal inferential screen. Model rankings are specific to this split and stated configurations. Hyperparameter searches resolve grid endpoints, but do not guarantee a continuous global optimum.

No additional course submission naming, upload, repository, or packaging rules appear in the supplied homework PDF. Only the supplied instructions are asserted here.

All 14 subquestions are answered in order. The item-by-item verification checklist is retained in results/completion_checklist.csv and the Notebook. No assignment items remain incomplete under the stated XGBoost interpretation.

Outstanding assignment items: none. A literal local-LASSO model tree is not claimed; the documented standard-XGBoost interpretation of 2(j) is used. Unprovided course submission rules cannot be verified. Final PDF layout and fresh-kernel execution are additionally checked by the packaging workflow.

References

- Supplied Homework5.pdf, Questions 1-2 and footnotes 1-5.

- Supplied diagnosis.names and diagnosis.data. J. Czerniak and H. Zarzycki, Application of rough sets in the presumptive diagnosis of urinary system diseases, ACS 2002, 2003, pp. 41-51.

- Supplied communities.names and communities.data, UCI Machine Learning Repository; source data: US Census 1990, LEMAS 1990, FBI Crime in the United States 1995. M. A. Redmond and A. Baveja, A Data-Driven Software Tool for Enabling Cooperative Information Sharing Among Police Departments, European Journal of Operational Research 141 (2002), 660-678.

- Matthew Mayo, Simplifying Decision Tree Interpretability with Python & Scikit-learn, KDnuggets (2017), supplied PDF. The path traversal here is a fresh implementation with label decoding and prediction assertions.

- Scikit-learn Ridge documentation and Lasso documentation, for objective conventions.

- XGBoost parameters and boosted-tree introduction, for booster and leaf-weight regularization semantics.